PROJECT · REQUIREMENTS · DOCUMENT

KhmerStay

Online Homestay & Guesthouse Booking Platform



"A B2C multi-vendor mobile e-commerce marketplace designed for Cambodia's local tourism sector."

PREPARED FOR

SETEC Institute

Lecturer · CHHAI Ousa

Subject · E-Commerce I

DRAFT FOR REVIEW

PREPARED BY

Men Manith
Sum Chansreyboth
Heng Ratanak
Phoem Sophanna
Khong Kimheng
Thy Somaneth
Hem Kimhong
Thoeun Chanthy

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1. Introduction

1.1 Purpose

This Project Requirements Document defines the functional and non-functional requirements of KhmerStay, a mobile-first online booking platform for homestays and guesthouses located throughout Cambodia. The document is intended for the development team, project supervisors, stakeholders, and evaluators, and serves as the authoritative reference for design, implementation, verification and acceptance of the system.

1.2 Product Overview

KhmerStay is a specialized digital marketplace designed specifically for the Cambodian local tourism sector. It links independent accommodation providers directly with domestic and international travelers through a convenient, secure and bilingual (Khmer / English) mobile application.

Key characteristics:

- B2C Multi-vendor Marketplace — a Business-to-Consumer model in which multiple independent property owners operate under a single trusted platform.
- Mobile E-commerce Concept — brings traditional offline bookings (Facebook chats, phone calls, walk-ins) into a convenient mobile application with real-time inventory.
- Connects Owners & Travelers — a seamless digital bridge between local homestays / guesthouses and tourists seeking authentic Cambodian stays.
- Trust & Transparency — verified vendors, real photos, clear pricing, ratings, reviews and instant booking confirmations.

1.3 Scope

The scope of the first release (v1.0) covers a fully functional cross-platform mobile application (Android & iOS) with an administrative web console. The system includes account management for three user roles, property and room management, search & discovery, an end-to-end booking workflow with online payment, reviews, notifications, reporting and admin oversight.

Out-of-scope items for v1.0 are captured in Section 12 (Future Roadmap).

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
B2C	Business-to-Consumer commercial model.
SRS	Software Requirements Specification.
FR / NFR	Functional / Non-Functional Requirement.
KHQR	Cambodia's national QR-code payment standard.
ABA PayWay	Payment gateway operated by ABA Bank Cambodia.
FCM	Firebase Cloud Messaging — push-notification service.
Vendor / Owner	A registered homestay or guesthouse operator.
Traveler	An end-user (domestic or international) booking a stay.
Admin	KhmerStay platform administrator.

1.5 References

- ISO/IEC/IEEE 29148:2018 — Requirements engineering.
- Ministry of Tourism of Cambodia — regulatory guidelines for accommodation providers.
- ABA PayWay Merchant Integration Guide.
- Firebase and Google Maps Platform documentation.

2. Background & Problem Statement

2.1 Current Situation

Booking a homestay or guesthouse in Cambodia today is still largely informal. Travelers usually rely on:

- Social media & chat apps — Facebook Pages, Messenger and Telegram used for enquiries and reservations.
- Phone calls & walk-ins — no digital record, no guaranteed availability.
- Scattered websites — a handful of independent sites with inconsistent quality, no unified search.

2.2 Problems for Travelers

- Difficulty comparing prices across properties.
- Unclear real-time room availability.
- Slow, chat-based communication with owners.
- Unverified property information and photos.
- No immediate confirmations or booking receipts.

2.3 Problems for Property Owners

- Limited online visibility beyond social media reach.
- Manual pen-and-paper records prone to error.
- Risk of duplicate or overlapping bookings.
- Difficulty managing multiple booking channels.
- Struggle to collect and showcase reviews.

2.4 The KhmerStay Solution

KhmerStay transforms the local tourism booking experience through:

Value	Description
One Mobile Marketplace	A single trusted app replaces fragmented Facebook pages and websites.
Effortless Booking	Search, compare, reserve and pay in a few taps.
Transparency	Verified listings, real photos, clear pricing, honest reviews.
Empowered Owners	Digital tools to manage listings, rooms, prices and reservations.
Secure Records	Cloud-based booking history — no more paper mistakes.
Promote Tourism	Showcases authentic Cambodian stays to a wider audience.

3. Objectives

3.1 Traveler Objectives

- Find accommodation quickly by destination, dates and guest count.
- Compare properties, rooms and prices side by side.
- Complete reservations securely with digital payment.
- View booking history and past receipts anytime.

3.2 Business-Owner Objectives

- Advertise properties to a nationwide audience.
- Digitally manage rooms, prices and availability.
- Receive and confirm reservations in real time.
- Monitor daily / monthly revenue through dashboards.

3.3 Platform Objectives

- Manage a healthy, trusted marketplace.
- Verify vendors to prevent fraud and low quality.
- Earn sustainable revenue through commission and premium services.
- Promote seasonal offers and support the Cambodian tourism industry.

4. Stakeholders

Stakeholder	Role & Interest
Traveler (End User)	Books stays; needs a fast, trustworthy and bilingual app.
Homestay / Guesthouse Owner	Lists properties and manages reservations; needs visibility and revenue tools.
Platform Administrator	KhmerStay operations team; verifies vendors, moderates content, monitors finances and analytics.
Payment Provider — ABA Bank	Processes online payments (KHQR, cards, bank transfer) via ABA PayWay.
Map & Location Provider — Google Maps	Supplies geolocation, routing and interactive map views.
Tourism Partners	Ministry of Tourism, local tour operators and marketing partners supporting the ecosystem.
Development Team	Designs, builds, tests and maintains the platform.

5. Business Model

KhmerStay operates as an intermediary B2C multi-vendor marketplace: the platform does NOT own accommodation inventory. Independent vendors publish listings; travelers book directly through the app; the platform earns a share of each successful transaction and offers optional value-added services.

5.1 Revenue Streams

Stream	Description	Priority
Booking Commission	Percentage fee (configurable, e.g. 8–12%) on every confirmed reservation.	Primary
Vendor Subscriptions	Optional monthly plans unlocking premium features and analytics.	Secondary
Featured Ads	Paid placement of properties on the home page / search top.	Secondary
Service Fees	Small transparent booking-service fee added to traveler's total.	Secondary
Partnerships	Revenue-sharing with tourism boards, tour operators, transport providers.	Growth

5.2 Price Calculation Formula

$$\text{Final Total} = (\text{RoomPrice} \times \text{Nights}) + \text{ServiceFee} + \text{Tax} - \text{Discount}$$

6. Functional Requirements

Functional requirements are grouped by user role. Each requirement has a unique identifier (FR-<role>-<n>) for traceability.

6.1 Traveler Requirements (FR-T)

6.1 Traveler Requirements (FR-T)

ID	Requirement	Priority
FR-T-01	The system shall allow a traveler to create an account using email or phone number with OTP verification.	Must
FR-T-02	The system shall allow social sign-in via Google and Facebook.	Should
FR-T-03	The system shall let the traveler manage a profile (name, photo, contact, ID for check-in, preferred language).	Must
FR-T-04	The system shall provide search by destination (province / city / landmark), check-in / check-out dates and guest count.	Must
FR-T-05	The system shall provide advanced filters: price range, property type, ratings, amenities (Wi-Fi, breakfast, A/C, parking).	Must
FR-T-06	The system shall display an interactive map view showing property pins around the searched location.	Must
FR-T-07	The system shall show property details: description, photo gallery, amenities, rules, price per night, cancellation policy, reviews.	Must
FR-T-08	The system shall let the traveler save properties to a Favorites list.	Should
FR-T-09	The system shall support the end-to-end booking workflow (see Section 7).	Must
FR-T-10	The system shall issue an instant booking confirmation with a unique reference number and QR code.	Must
FR-T-11	The system shall let the traveler view current, upcoming and past bookings and download receipts (PDF).	Must
FR-T-12	The system shall let the traveler cancel a booking subject to the vendor's cancellation policy.	Must
FR-T-13	The system shall let the traveler rate a stay (1–5 stars) and leave a review after check-out.	Must
FR-T-14	The system shall send push, email and/or SMS notifications for booking status changes, reminders and promotions.	Must

6.2 Owner / Vendor Requirements (FR-O)

ID	Requirement	Priority
FR-O-01	The system shall allow owners to register a business account and submit verification documents (national ID, business license).	Must

ID	Requirement	Priority
FR-O-02	The system shall let owners create and manage property listings (name, address, description, location on map, amenities, house rules).	Must
FR-O-03	The system shall let owners add multiple room types with capacity, photos, price and count of units.	Must
FR-O-04	The system shall let owners upload up to N high-resolution images per property / room.	Must
FR-O-05	The system shall let owners set pricing including seasonal / weekend surcharges and promotional discounts.	Should
FR-O-06	The system shall provide a real-time calendar showing reservations and blocked dates.	Must
FR-O-07	The system shall let owners accept, reject or auto-confirm reservations.	Must
FR-O-08	The system shall provide an in-app messaging channel between owner and traveler (post-booking, v1.0 basic).	Should
FR-O-09	The system shall provide an earnings dashboard (daily, weekly, monthly revenue and payouts).	Must
FR-O-10	The system shall let owners respond publicly to guest reviews.	Must

6.3 Administrator Requirements (FR-A)

ID	Requirement	Priority
FR-A-01	The system shall provide a secure admin console with role-based access control and 2-factor authentication.	Must
FR-A-02	The system shall let admins verify, suspend or ban vendors and users.	Must
FR-A-03	The system shall let admins moderate property listings, photos and reviews.	Must
FR-A-04	The system shall let admins monitor all payments and refunds.	Must
FR-A-05	The system shall let admins configure the commission rate globally or per vendor tier.	Must
FR-A-06	The system shall let admins create platform-wide promotions and discount codes.	Should
FR-A-07	The system shall provide analytics dashboards (GMV, bookings, top destinations, cancellation rate, top vendors).	Must
FR-A-08	The system shall provide exportable reports (CSV / PDF) for finance and compliance.	Should

7. Booking Workflow

The end-to-end reservation flow consists of nine steps. The system shall enforce this sequence and validate inputs at each step.

Step	Action	System Behaviour
1	Search	Traveler enters destination, dates and guest count.
2	Select Property	Traveler chooses a property from the results / map.
3	Choose Room	Available room types shown for the chosen dates.
4	Select Dates	Interactive calendar confirms availability.
5	Enter Guest Info	Traveler enters names, contact and ID for check-in.
6	Review Price	Price breakdown shown per formula in §5.2.
7	Select Payment	Traveler picks KHQR / ABA PayWay / card / bank transfer.
8	Confirm	System reserves inventory and calls the payment gateway.
9	Receive Reference	Booking reference, QR code and email/SMS receipt issued.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All traffic shall use HTTPS/TLS 1.2+; passwords hashed with bcrypt; PII encrypted at rest; PCI-DSS-compliant payment via ABA PayWay (no card data stored in KhmerStay).
NFR-02	Privacy	The system shall comply with Cambodian data-protection guidelines and provide a clear privacy policy and user-controlled data deletion.
NFR-03	Performance	Search results shall return within 2 seconds for 95% of queries; app cold-start under 3 seconds on mid-range devices.
NFR-04	Scalability	The backend shall support at least 10,000 concurrent users and scale horizontally via Firebase / cloud auto-scaling.
NFR-05	Availability	Target monthly uptime ≥ 99.5% with automated daily backups.
NFR-06	Usability / UX	Mobile-first responsive UI following Material Design 3; task completion (search → booking) achievable in ≤ 6 taps for a returning user.
NFR-07	Localization	Full bilingual support: Khmer (default) and English, switchable at runtime; currency displayed in KHR and USD.
NFR-08	Accessibility	Meets WCAG 2.1 Level AA where applicable (contrast, font scaling, TalkBack / VoiceOver labels).
NFR-09	Reliability	The booking workflow shall be transactional — no double-booking even under concurrent requests.
NFR-10	Maintainability	Code follows Flutter / Dart style guides; ≥ 70% unit-test coverage on business logic.
NFR-11	Portability	Single Flutter codebase builds for Android (API 24+) and iOS (13+).
NFR-12	Auditability	All admin, payment and booking actions are logged with timestamp, actor and IP.

9. Technology Stack

Layer	Technology	Purpose
Mobile Front-end	Flutter + Dart	Cross-platform mobile UI for Android and iOS.
State Management	Provider / Riverpod / BLoC	Predictable state and separation of concerns.
Authentication	Firebase Authentication	Email/phone/OTP and social sign-in.
Database	Supabase (PostgreSQL)	Managed relational database with built-in connection pooling (Supavisor) and JSON/relational querying.
File Storage	Firebase Storage	Property and room photos, documents, receipts.
Notifications	Firebase Cloud Messaging (FCM)	Push notifications on Android / iOS.
Maps & Location	Google Maps Platform	Map view, geocoding, distance and directions.
Payments	ABA PayWay (KHQR, cards, bank)	Secure online payment processing in Cambodia.
Backend / APIs	Golang (Gin)	SSR/SSG administrative dashboard for property approvals, analytics, and platform controls.
Admin Console	Next.js (React / TypeScript)	Browser-based administration and reporting.
Analytics	Firebase Analytics + Crashlytics	Usage insights and crash reporting.
CI / CD	GitHub Actions + Codemagic	Automated mobile builds (Codemagic) and automated backend/admin deployments (Render & Vercel).

10. High-Level System Architecture

The KhmerStay system follows a client–server architecture with a cloud-hosted backend and third-party service integrations. The Flutter mobile client communicates with Firebase services and a REST API layer that orchestrates business logic and external gateways (ABA PayWay, Google Maps).

Main components:

1. Mobile Client (Flutter) — Traveler and Owner apps sharing the same codebase with role-based UI.

2. Admin Web Console — browser-based dashboard for the platform team.
3. Firebase Backend — Auth, Firestore, Storage, FCM, Functions.
4. Payment Gateway — ABA PayWay via secure REST integration.
5. Map Service — Google Maps SDK / Platform APIs.
6. Notification Service — FCM plus email/SMS providers.

11. Assumptions, Dependencies and Constraints

11.1 Assumptions

- Users have access to a smartphone with mobile Internet.
- Vendors are willing to digitize their operations and provide legal documents for verification.
- Merchant accounts with ABA Bank and Google Cloud are available to the project.

11.2 Dependencies

- Firebase and Google Maps service availability and quotas.
- ABA PayWay merchant integration approval.
- Google Play Store and Apple App Store review policies.

11.3 Constraints

- Project timeline: one academic semester (v1.0).
- Team size and budget as defined by the course.
- Compliance with Cambodian tourism and payment regulations.

12. Future Roadmap (Post v1.0)

- AI-powered personalized recommendations for travelers.
- Live KHQR generation for direct in-app payment.
- Full in-app chat with photo sharing and translation.
- Loyalty and rewards program for repeat travelers.
- Digital self check-in / check-out with QR code.
- Additional regional-language support (Chinese, French, Korean).
- Expansion into tour, activity and transport booking.

13. Acceptance Criteria

The KhmerStay v1.0 release will be considered acceptable when:

1. All Must-priority functional requirements in Section 6 are implemented, tested and demonstrated end-to-end.
2. The booking workflow (Section 7) completes successfully using a real ABA PayWay sandbox transaction.
3. All non-functional requirements in Section 8 are met or have documented, agreed exceptions.
4. The application passes usability testing with at least 10 representative travelers and 5 vendors.
5. The admin console demonstrates verification, moderation and reporting workflows.
6. Documentation (user guide, admin guide, deployment guide) is delivered.

14. Approval and Sign-off

By signing below, the parties acknowledge that they have reviewed and approve the requirements described in this document as the baseline for the KhmerStay project.

Role	Name	Signature	Date
Project Supervisor / Lecturer			
Project Manager			
Lead Developer			
Client / Stakeholder Representative			

15. Implemented Functions by Project Module

This section lists the functions present in the current KhmerStay source code as reviewed on 22 October 2026. Functions are grouped by project and module. Status labels identify functions that work through the local Go API, functions that need external credentials, and screens that remain part of the school-project demonstration.

15.1 Go API

The backend provides the REST API, authorization, SQLite persistence, local image storage, email OTP delivery, and conditional Bakong KHQR integration.

Module	Main feature	Function list	Current status
System and runtime	**Service startup and health**	Start the Go HTTP service, load environment settings, expose the health check, apply CORS rules, decode requests, and return consistent JSON responses and errors.	Implemented
Data storage	**SQLite persistence**	Create and seed tables for users, sessions, OTP records, properties, rooms, bookings, payments, favorites, conversations, messages, reviews, notifications, blocked dates, promotions, settings, and audit logs.	Implemented
Authentication	**Accounts and sessions**	Register traveler or owner accounts, sign in with email and password, create seven-day sessions, return the signed-in user, sign out, hash passwords with bcrypt, and enforce role-based access.	Implemented
Email OTP	**Verification and recovery**	Generate random six-digit codes, store only hashed codes, enforce expiry, resend delay and attempt limits, verify registration or email sign-in, and reset passwords through SMTP delivery.	Implemented; SMTP account required for external email
Property catalog	**Browse and filter stays**	List published properties for travelers; expose owner and administrator records according to access; filter by search text, province, category and status; return property details.	Implemented
Property management	**Listings and rooms**	Create owner listings, edit permitted property fields, store room types and inventory, submit a listing for review, and let administrators publish, reject, pause or restore listings.	Implemented
Image storage	**Property photo upload**	Accept authenticated PNG, JPEG and WebP uploads up to 8 MB, save files in local storage, and serve uploaded images from the API.	Implemented
Booking	**Reservation creation**	Validate dates and guest capacity, check blocked dates and room inventory, calculate the stay total, create a booking reference, create its payment record, and open the related conversation.	Implemented
Booking	**Reservation lifecycle**	List role-scoped bookings, return booking details, enforce traveler, owner and administrator permissions, and validate cancellation, confirmation, check-in, completion, decline and no-show transitions.	Implemented
KHQR payment	**QR generation and verification**	Generate a transaction-specific KHQR payload and QR image, store its MD5 and expiry, check payment through the Bakong API, match account, currency and amount, then confirm the booking after verification.	Implemented; Bakong account and token required
KHQR payment	**Local payment demonstration**	Allow a traveler to confirm a pending payment in development when Bakong is not configured; disable this route outside development and when a receiving account is configured.	Development fallback
Favorites	**Saved properties**	List, add and remove a traveler's favorite published properties.	Implemented

Module	Main feature	Function list	Current status
Messaging	**Owner and traveler conversations**	List authorized conversations, retrieve messages, create a conversation when needed, send text messages, enforce participant access, and notify the recipient.	Implemented
Notifications	**In-app updates**	Create booking, payment, message and review notifications; list notifications for the signed-in user; mark an individual notification as read.	Implemented
Reviews	**Guest reviews and owner responses**	List reviews by property, allow a traveler with a completed stay to submit a 1-5 rating and comment, and allow the property owner or administrator to publish a response.	Implemented
Availability	**Blocked date management**	List blocked dates by owner or property and allow an authorized owner or administrator to block or unblock a valid calendar date.	Implemented
Administration	**Operations and configuration**	Return role-based dashboard totals, list users, suspend or reactivate accounts, read and update approved platform settings, create promotions, and record audit events.	Implemented

15.2 Flutter Mobile Application

The Flutter application contains traveler and owner experiences in one codebase and connects its primary data flows to the Go API.

Module	Main feature	Function list	Current status
Application startup	**Launch and navigation**	Display splash, onboarding, preference setup, role selection and welcome screens; route users into traveler or owner navigation with animated page transitions.	Implemented
Preferences	**Language and currency**	Switch between Khmer and English, select USD or KHR display, and retain onboarding, language, currency and role preferences locally.	Implemented locally
Authentication	**Account access**	Sign in with password, register a traveler or owner, request and verify an email OTP, sign in by OTP, reset a password, restore the API session, sign out, and require sign-in when switching roles.	API integrated
Traveler discovery	**Explore and search**	Load published properties from the API, browse categories, search destinations, filter results, and open stay details.	API integrated
Traveler discovery	**Property information**	Show property description, room choices, prices, amenities, photo gallery, host profile and guest reviews; load uploaded images from the local API.	API integrated
Favorites	**Wishlist**	Add or remove published properties from favorites and load the signed-in traveler's saved list.	API integrated
Booking	**Reservation flow**	Select a room, choose future dates, set guest count, enter guest details, review the API-aligned total, accept the booking conditions, and create a reservation.	API integrated
Payment	**KHQR checkout**	Display the booking amount and payment state, show the generated KHQR image when configured, request Bakong payment verification, or use the labeled local demonstration confirmation.	API integrated; credentials required for real payment
Trips	**Booking history and status**	Load traveler bookings, group active and past trips, open booking details, and request an allowed cancellation.	API integrated

Module	Main feature	Function list	Current status
Messaging	**Inbox and chat**	Load conversations and messages for the signed-in account, open a booking conversation, send text messages, and refresh the conversation state.	API integrated
Reviews	**Ratings and responses**	Load property reviews, allow eligible travelers to submit a rating and comment, and let owners read reviews and save or edit a public response.	API integrated
Notifications	**Updates**	Load API notifications, display unread state, mark a notification as read, and open notification-related content.	API integrated
Traveler account	**Profile and preferences**	Display traveler account options, update app language and currency, access notifications and information screens, switch to owner sign-in, and sign out.	Partly local; authentication is API integrated
Owner dashboard	**Operational summary**	Show owner properties, bookings and revenue summaries using the application state loaded from the API.	API-backed summary
Owner bookings	**Reservation operations**	List owner bookings, inspect guest and stay details, and apply allowed confirmation, decline, check-in and completion transitions.	API integrated
Owner properties	**Listing creation and photos**	List owner properties, open listing details, create or edit a property, upload a photo, and submit listing information to the API.	API integrated
Owner inventory	**Rooms and pricing**	Load each property's room types, add or edit room name, bed, capacity, price and inventory, and save nightly price changes.	API integrated
Owner availability	**Calendar**	Select an owner property, view current and future dates, see bookings that cover a selected date, and block or unblock availability.	API integrated
Owner secondary screens	**Earnings verification and help**	Present earnings, verification and help content for the school demonstration without external payout, identity verification or support services.	Demonstration UI

15.3 Next.js Website

The website provides owner and administrator portals. Core records are loaded from the Go API, while some external-service and secondary administration workflows remain demonstration screens.

Module	Main feature	Function list	Current status
Authentication	**Owner and administrator access**	Sign in through the Go API, store the session, validate the session with the current-user endpoint, enforce owner or administrator portal access, reset passwords by email OTP, and sign out.	API integrated
Portal shell	**Navigation and preferences**	Provide responsive owner and administrator navigation, global search, Khmer or English labels, USD or KHR display, notification access, role switching through sign-in, and session-aware account menus.	Implemented
Shared data	**API hydration**	Load properties, bookings, notifications, conversations, promotions, blocked dates, settings and users at portal startup and populate shared dashboard state.	API integrated
Owner dashboard	**Portfolio overview**	Display counts for properties and bookings, payment totals, recent reservations and operational shortcuts from API-loaded data.	API-backed summary
Owner properties	**Listing management**	List owner properties, open details, create or edit listing fields, upload a property photo, submit new listings for review, and change allowed listing status.	API integrated
Owner rooms	**Inventory management**	Load room types for a property, add or edit room name, bed, capacity, nightly price and inventory, then save the room collection.	API integrated
Owner pricing	**Nightly rates**	Load property room data, update room prices, and persist the change through the property API.	API integrated
Owner calendar	**Blocked dates**	Display booking and blocked-date information and save availability changes through the blocked-date API.	API integrated
Owner bookings	**Reservation management**	List bookings, open a dynamic booking detail page, view payment and guest details, and apply valid owner status transitions.	API integrated
Owner messages	**Guest communication**	List booking conversations, load message history, select a conversation and send a new message.	API integrated
Owner reviews	**Public responses**	Load reviews for the owner's properties, search feedback, calculate the current average rating, and create or edit a public response.	API integrated
Owner notifications	**Operational alerts**	Display API notifications and persist read status through the notification endpoint.	API integrated
Owner reporting	**Earnings and analytics**	Summarize API-loaded booking totals and six-month booking and revenue trends; show supporting school-project charts.	Partly API backed
Owner account	**Verification payouts and profile**	Provide account, verification and payout screens for demonstrating the intended workflow without an external identity provider, bank payout service or account-profile update endpoint.	Demonstration UI
Administrator dashboard	**Marketplace overview**	Display API-derived property, booking and revenue totals with shortcuts to administrative work areas.	API-backed summary
User administration	**Traveler and owner accounts**	List users, filter travelers or owners, and suspend or reactivate an account with administrator authorization.	API integrated
Property moderation	**Listing review**	List all properties for administrators, inspect listing details, and publish, reject, pause or restore a property through the status endpoint.	API integrated
Booking administration	**Reservation oversight**	List all bookings, open booking details, inspect payment state, and apply administrator-allowed status changes.	API integrated
Platform management	**Promotions and settings**	Load and create promotion codes; read and update commission, maintenance, instant-booking and payout settings.	API integrated
Administrator secondary screens	**Finance moderation content support and audit**	Present finance, verification, moderation, catalog, content, support and audit-log workspaces; these screens currently combine API-loaded records with demonstration data and local actions.	Partial demonstration